TCP/IP-Based Smart Irrigation System for Real-Time Soil Moisture Monitoring and Remote Farm Management

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

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to the award of the degree of

BACHELOR OF ENGINEERING IN

ELECTRONICS AND COMMUNICATION ENGINEERING

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "TCP/IP-Based Smart Irrigation System for Real-Time Soil Moisture Monitoring and Remote Farm Management" has been carried out by Sudhakar V (192412125) under the supervision of Dr. SENTHAMILSELVAN R and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Electronics and Communication Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I,Sudhakar V (192412125) of the ECE Department, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “TCP/IP-Based Smart Irrigation System for Real-Time Soil Moisture Monitoring and Remote Farm Management” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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Individual Contribution Statement

Student	Specific Responsibilities	Design & Development	Testing & Analysis	Report Contribution	Approx. %
Sudhakar V	Conceptualized the smart irrigation system, selected the sensor-microcontroller architecture, and managed end-to-end execution from hardware setup to network communication.	Designed the circuit integrating soil moisture, temperature, and humidity sensors with a microcontroller (e.g., ESP32/Arduino with Ethernet/Wi-Fi module), and developed the TCP/IP-based communication protocol for remote data transmission and control of the water pump/solenoid valve.	Tested sensor accuracy under varying soil conditions, verified reliable TCP/IP packet transmission between the client and server, and analyzed system response time and irrigation efficiency compared to manual watering.	Wrote the Introduction, System Architecture, and Implementation sections; prepared circuit diagrams, network flowcharts, and result graphs; compiled and finalized the complete report.	45%
Total					100%

ABSTRACT

Agricultural irrigation is often operated using fixed schedules or manual observation, which can lead to over-irrigation, under-irrigation, unnecessary pumping energy, and delayed response to changing soil conditions. The engineering problem addressed in this project is the design of a connected irrigation system that can continuously observe soil moisture, make a local irrigation decision, communicate operational information over a TCP/IP network, and allow a remote user to monitor and manage a farm zone.

The proposed system uses a soil-moisture sensing node built around an ESP32-class microcontroller, a relay or motor-driver interface, and a water pump or solenoid valve. The controller samples soil moisture at a defined interval, compares the measured value with configurable irrigation thresholds, and switches the irrigation actuator accordingly. TCP/IP connectivity provides a bidirectional communication path to a web/mobile dashboard or local server, where the user can observe moisture, pump status, connectivity status, and historical measurements. Safety logic is included to prevent unintended continuous pumping and to provide fail-safe behavior when sensor or network faults are detected.

The design methodology covers requirements definition, architecture selection, hardware/software integration, communication design, verification planning, risk assessment, sustainability, and project management. Because measured field data were not supplied with the project brief, numerical performance values in this report are defined as engineering acceptance targets and example validation criteria rather than claims of completed field measurements. The system is intended to demonstrate a practical closed-loop IoT irrigation architecture that reduces dependence on manual operation while retaining local control when network connectivity is unavailable.

The principal engineering contribution is the integration of real-time sensing, threshold-based control, TCP/IP communication, remote supervision, and safety mechanisms into one low-cost prototype architecture. The resulting design provides a scalable foundation for multi-zone irrigation and future enhancements such as predictive irrigation, weather integration, anomaly detection, and secure cloud deployment.

Keywords: Smart irrigation; Soil moisture monitoring; TCP/IP; Internet of Things; ESP32; Remote farm management.

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol / Abbreviation	Description	Unit
IoT	Internet of Things	—
TCP	Transmission Control Protocol	—
IP	Internet Protocol	—
HTTP	Hypertext Transfer Protocol	—
ESP32	Wi-Fi-enabled microcontroller family	—
ADC	Analog-to-Digital Converter	counts
RH	Relative Humidity	%
SM	Soil Moisture	% (mapped)
GPIO	General-Purpose Input/Output	—
API	Application Programming Interface	—
MQTT	Message Queuing Telemetry Transport	—
V	Voltage	V
I	Current	A
P	Power	W
Δt	Sampling interval	s

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Water is a critical input to agriculture, but the amount required by a crop varies with soil condition, crop stage, weather, irrigation method, and time of day. A fixed irrigation schedule cannot directly observe the moisture state of the root zone. Manual irrigation also depends on the availability and judgement of the operator. These limitations motivate sensing and automation at the field level.

An IoT-based irrigation system combines physical sensing, embedded computation, communication, and actuation. In the proposed architecture, a soil-moisture sensor provides the primary feedback variable. An ESP32-class controller processes the reading and operates a pump or valve through an electrically isolated switching stage. A TCP/IP network transports telemetry and control information to a remote dashboard. Local control is retained so that irrigation does not depend entirely on the availability of a remote server.

The system is designed as a modular prototype suitable for one irrigation zone. The same architecture can be extended to multiple zones by assigning additional sensing/actuation channels and addressing each node over the IP network.

1.2 Need for the Project

- Reduce unnecessary irrigation caused by timer-only operation or human estimation.
- Provide continuous visibility of soil moisture and irrigation status to a remote operator.
- Respond to changing soil conditions using feedback rather than a fixed schedule alone.
- Reduce the need for a farmer to be physically present at the irrigation control point.
- Provide local fail-safe behavior when sensors, network links, or remote services are unavailable.
- Create a low-cost, scalable platform for future multi-zone and data-driven irrigation control.

The affected stakeholders include farmers, farm supervisors, irrigation technicians, agricultural automation developers, and end users responsible for water and energy management. From an engineering perspective, the challenge is not only to measure moisture but to maintain reliable interaction between a wet outdoor environment, low-voltage electronics, a motor/valve load, and a packet-switched communication network.

1.3 Problem Statement

Design and implement a TCP/IP-connected smart irrigation prototype that continuously monitors soil moisture, determines when irrigation is required using configurable control thresholds, actuates a pump or valve safely, and provides real-time remote monitoring and management. The system shall balance water-use efficiency, response time, communication reliability, electrical safety, environmental exposure, cost, and ease of operation while maintaining local irrigation control during temporary network or server failure.

1.4 Project Aim

To develop and validate a reliable IoT-based irrigation system that uses real-time soil-moisture feedback and TCP/IP communication to automate irrigation and enable remote farm management.

1.5 Project Objectives

1. Design a complete sensing, processing, communication and actuation architecture for one irrigation zone.

2. Interface a soil-moisture sensor with an ESP32-class controller and establish a repeatable moisture-to-percentage mapping.

3. Implement threshold/hysteresis-based irrigation control to reduce rapid pump switching.

4. Transmit moisture, irrigation state and device-health information over a TCP/IP network.

5. Provide a remote dashboard/control interface for monitoring and authorized manual intervention.

6. Define and execute verification tests for sensing, actuation, network communication, fault handling and safety.

7. Evaluate design trade-offs, sustainability, security, risks, cost and future scalability.

1.6 Scope

Included: one-zone soil-moisture sensing; embedded controller; relay/MOSFET-based pump or valve switching; TCP/IP communication over Wi-Fi/Ethernet gateway; local status indication; remote monitoring; configurable thresholds; basic fault handling; testing methodology; risk and security analysis.

Excluded: agronomic optimization for a specific crop, large-scale hydraulic design, commercial-grade cloud infrastructure, autonomous weather forecasting, certified industrial safety certification, and multi-season field trials. These are future extensions or separate engineering activities.

Assumptions: the irrigation source and pump/valve are available; the sensor is installed in a representative root-zone location; network coverage exists at the controller location; the user can configure appropriate thresholds; and the prototype is operated within the electrical ratings of its components.

Boundary conditions: outdoor deployment requires suitable enclosure and cable protection; mains-powered pumps must be isolated from low-voltage electronics; irrigation must stop on detected critical faults; and the network is considered an untrusted communication medium.

1.7 Expected Engineering Outcome

The intended outcome is a working embedded/IoT prototype comprising a soil-moisture sensing node, local control firmware, safe actuator interface, TCP/IP communication service, and remote monitoring/control interface. The system should demonstrate closed-loop irrigation behavior, visible device status, and documented verification criteria. Numerical field-performance claims are intentionally not presented as completed results without measured project data.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The review was structured around five technology areas relevant to the proposed system: soil-moisture sensing, embedded IoT controllers, TCP/IP networking, automated irrigation control, and remote farm monitoring. Representative academic literature, established protocols, manufacturer documentation, and standards were considered to identify common architectures and unresolved engineering issues. The review emphasizes system-level relevance rather than treating sensing, communication, and actuation as isolated technologies.

2.2 Review of Existing Work

Soil-moisture sensing is commonly implemented using resistive or capacitive probes. Resistive probes are inexpensive but may be affected by electrode corrosion and changes in soil conductivity. Capacitive sensing reduces direct electrode contact with the soil and is therefore attractive for longer-term prototypes, although calibration remains dependent on soil type and installation.

Microcontrollers with integrated wireless connectivity have become practical platforms for agricultural IoT prototypes because they combine ADC/GPIO interfaces with network connectivity. ESP32-class devices are suitable for this project because a single controller can read sensors, execute control logic, maintain TCP/IP connectivity and operate a switching interface.

TCP/IP provides a mature networking foundation for remote farm management. HTTP/REST can simplify dashboard integration, while MQTT can reduce messaging overhead for telemetry-oriented systems. The present design uses TCP/IP as the transport/network foundation and keeps the application protocol modular so that HTTP or MQTT can be selected during implementation.

Automated irrigation systems generally use timer-based, threshold-based, or model-based control. Timer control is simple but does not respond to actual soil conditions. Threshold control introduces feedback with modest computational requirements. More advanced approaches can incorporate weather, evapotranspiration, crop models or machine learning, but they require additional data and validation.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Manual irrigation	Low initial cost; simple	Labour intensive; inconsistent timing; no continuous sensing	Baseline problem to automate
Fixed timer irrigation	Simple; predictable	Ignores actual soil condition; can over/under-water	Used only as a fallback/reference
Standalone moisture-triggered controller	Local feedback; no internet dependency	Limited remote visibility and management	Core local-control concept
Wi-Fi/TCP-IP IoT irrigation	Remote monitoring; configurable; scalable	Network/security dependency; more software complexity	Primary proposed architecture
Cloud/ML irrigation	Can use historical/weather data	Higher complexity; data and model requirements	Future enhancement

2.4 Research / Engineering Gap

A recurring gap between simple moisture-triggered controllers and connected agricultural platforms is the need to integrate reliable local control with remote visibility without making irrigation dependent on a remote service. A network outage should not automatically disable the basic safety and irrigation logic. In addition, prototype systems often under-specify electrical isolation, fault handling, cybersecurity, and acceptance criteria.

The proposed work addresses this system-level gap by separating local control from remote supervision: moisture-based decisions occur at the controller, while TCP/IP communication is used for telemetry and authorized remote commands. The design also explicitly considers sensor faults, actuator timeouts, network loss, authentication, and safe power interfacing.

2.5 Proposed Contribution

The proposed contribution is a practical one-zone reference architecture combining real-time soil-moisture sensing, hysteresis-based local irrigation control, TCP/IP telemetry, remote farm management, and documented safety/security mechanisms. The architecture is intentionally modular so that future versions can add multiple zones, weather inputs, historical analytics, and predictive control without redesigning the fundamental sensing-to-actuation chain.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Farmer / Operator	View current soil moisture and irrigation status remotely.
Farmer / Operator	Start/stop irrigation manually when authorized, with safety limits.
Farm Supervisor	Use configurable thresholds rather than fixed, hidden settings.
Maintenance Technician	Identify sensor, network and actuator faults.
Project Assessor	Demonstrate measurable sensing, communication and control behavior.
Environment / Community	Avoid unnecessary water use and unsafe discharge/operation.

Functional & Non-Functional Requirements

Requirement Type	Requirement	Measurable Target
Functional	Measure soil moisture periodically	Configurable 5–60 s interval
Functional	Automatic irrigation control	ON below lower threshold; OFF above upper threshold
Functional	Remote telemetry	Current value and actuator state visible to client
Functional	Manual remote control	Authorized command accepted/rejected deterministically
Functional	Fault handling	Stop irrigation on critical sensor/actuator fault
Non-Functional	Control stability	Hysteresis prevents repeated switching near threshold
Non-Functional	Network resilience	Local control continues during temporary TCP/IP loss
Non-Functional	Electrical safety	Low-voltage control isolated from pump load
Non-Functional	Security	Authenticated control path; no unauthenticated actuation
Non-Functional	Maintainability	Modular firmware and documented pin/protocol mapping

3.2 Constraints & Applicable Standards

Standard / Code	Requirement	How Applied in This Project
IEC 60335-1 (principles)	Electrical safety principles for appliances	Used as design guidance for isolation, enclosure and protection; prototype is not claimed certified.
IEC 60529	Ingress protection classification	Guides enclosure selection for outdoor deployment; final IP rating depends on actual enclosure.
ISO/IEC 27001 principles	Information-security risk management	Used conceptually for access control, asset/risk identification and secure operation.
OWASP IoT guidance	Common IoT security practices	Applied through unique credentials, least privilege, secure updates and reduced exposed services.
IEEE 802.11	Wi-Fi communication	Used where ESP32 communicates through a Wi-Fi LAN.
TCP/IP (IETF RFC 9293 for TCP; RFC 791/related IP specifications)	Reliable network transport	Used as the communication foundation for remote monitoring/control.

The listed standards and guidance are design references. The student prototype should not be represented as formally certified against a standard unless laboratory certification or a compliant assessment has actually been completed.

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Moisture measurement interval	10 (configurable)	s	High
Moisture operating range	0–100 mapped scale	%	High
Automatic ON threshold	Example 35	%	High
Automatic OFF threshold	Example 55	%	High
Hysteresis	20	percentage points	High
Control supply	3.3/5 depending on module	V	High
Network transport	TCP/IP over local IP network	—	High
Dashboard refresh	≤5 for normal LAN operation	s	Medium
Maximum continuous pump run	Configurable safety timeout	min	High
Actuator isolation	Relay/driver isolation appropriate to load	—	High

3.4 Alternative Solutions & Evaluation

Alternative 1: Timer-based irrigation with no soil sensing. This option is simple and inexpensive but cannot adapt to actual soil condition and offers limited remote intelligence.

Alternative 2: Local moisture-triggered controller without networking. This provides useful feedback control and robust local operation but does not provide remote monitoring or farm management.

Alternative 3: TCP/IP IoT controller with local feedback and remote supervision. This option adds communication and software complexity but provides the best balance between autonomous local control, visibility, and future scalability.

Criterion	Weight	Alt. 1 Timer	Alt. 2 Local Sensor	Alt. 3 TCP/IP IoT
Performance	0.25	2	4	5
Cost	0.15	5	4	3
Safety	0.2	3	4	5
Sustainability	0.15	2	4	5
Reliability	0.15	4	5	4
Remote management	0.1	1	1	5
Weighted score	1.00	2.75	3.85	4.65

3.5 Selected Design & Detailed Design

Alternative 3 is selected because its weighted evaluation provides the strongest overall balance for the stated project requirements. Although it has higher software and integration complexity than a standalone controller, local threshold control limits the impact of network failure while TCP/IP connectivity enables the principal remote-management function.

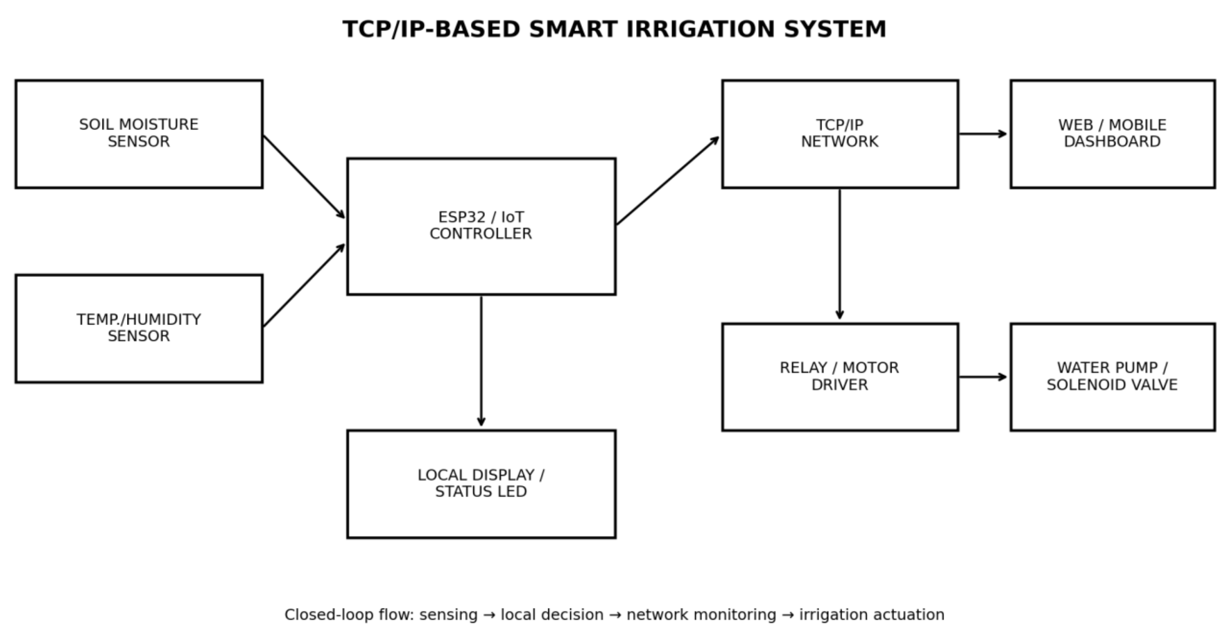

Figure 3.1: System architecture and information flow

The soil-moisture sensor is sampled by the controller's ADC or digital interface. The controller converts the raw reading to a calibrated percentage or normalized scale, applies filtering and hysteresis, and updates the actuator state. Telemetry is periodically transmitted through the TCP/IP stack to a dashboard/server. Remote commands are validated by the controller before any actuation is permitted.

Key calculation: for a DC pump, electrical input power can be approximated by $P = V \times I$. For example, a 12 V pump drawing 1.5 A requires approximately 18 W electrical input before accounting for driver losses. The switching device, wiring, supply and protection therefore need ratings above the expected steady-state load and appropriate margin for startup current.

For a threshold controller, if M is mapped soil moisture, T_ON is the lower threshold and T_OFF is the upper threshold, then: if $M \leq T\_ON$, irrigation is enabled; if $M \geq T\_OFF$, irrigation is disabled; otherwise the previous state is retained. This hysteresis prevents rapid toggling when M fluctuates around a single threshold.

Systems approach: the sensing subsystem provides the feedback variable to the controller; the controller is the decision and coordination layer; the actuator subsystem converts the decision into water flow; and the TCP/IP subsystem provides remote observation and authorized intervention. A failure in the network must not remove the controller's local safety function. A failure in the sensor must inhibit automatic watering rather than cause indefinite pumping.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Sensor type	Resistive vs capacitive	Resistive is cheaper	Corrosion and soil-contact effects	Prefer capacitive for prototype longevity; calibrate to soil.
Control algorithm	Single threshold vs hysteresis	Single threshold is simpler	Can chatter near threshold	Use hysteresis for stable actuation.
Connectivity	Wi-Fi/TCP-IP vs standalone	TCP/IP enables remote management	Network dependency	Use TCP/IP while retaining local control.
Remote protocol	HTTP vs MQTT	HTTP is easy to debug; MQTT suits telemetry	Different server/client complexity	Use modular application layer; select based on deployment.
Actuation	Relay vs transistor/MOSFET	Relay isolates load; MOSFET is efficient for DC	Relay wear; MOSFET needs correct protection	Use suitable driver matched to pump/valve and voltage.

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Reduce unnecessary pumping and water use	Feedback control vs timer operation	Moisture-based control selected; thresholds configurable.

Ethics	Avoid misleading performance claims	No measured field dataset supplied	Results are labelled as targets/proposed validation until measured.
Health & Safety	Pump/mains isolation; outdoor exposure	Load current, enclosure, cable routing	Separate low-voltage logic from load and use protection.
Security	Prevent unauthorized irrigation commands	Network is treated as untrusted	Authentication, least privilege and local command validation included.

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Sensor gives invalid value	Medium	High	High	Range checking, timeout, disable automatic irrigation	Medium
Network unavailable	Medium	Medium	Medium	Local control continues; buffer status	Low
Pump runs continuously	Low/Medium	High	High	Maximum run-time watchdog and fault shutdown	Low
Water leak / pipe failure	Medium	High	High	Flow/pressure sensing as future improvement; runtime limit	Medium
Unauthorized remote command	Medium	High	High	Authentication, authorization, network segmentation, secure transport where supported	Low/Medium
Electrical shock / short circuit	Low	High	High	Isolation, fusing, enclosure, correct ratings	Low
Power failure	Medium	Medium	Medium	Safe restart state; irrigation resumes only after valid sensing	Low/Medium

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The development process follows an incremental engineering workflow: define requirements, prototype sensing, validate actuator switching, implement local control, establish TCP/IP communication, integrate remote monitoring, then verify the complete system under normal and fault conditions. Each subsystem is tested independently before integration so that failures can be localized. Configuration values such as thresholds and sampling interval are kept adjustable rather than hard-coded into the architecture.

Resource	Purpose
ESP32-class development board	Embedded processing, Wi-Fi/TCP-IP and GPIO/ADC
Capacitive soil-moisture sensor	Primary feedback measurement
Relay module / MOSFET driver	Electrical interface to pump/valve
DC pump or solenoid valve	Irrigation actuation
Regulated power supply	Logic and actuator power
Breadboard/PCB, wires, connectors	Prototype interconnection
Laptop / development workstation	Firmware, dashboard and testing
Local Wi-Fi router	TCP/IP network path
Dashboard/server software	Remote monitoring and control

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

Hardware Implementation

The sensor output is connected to an ADC-capable input or digital interface as appropriate to the selected sensor. The controller operates a driver stage rather than directly powering the pump. For a DC inductive load, flyback protection is required when a transistor/MOSFET stage is used; when a relay module is used, the module must be rated for the pump's voltage and current, including startup conditions.

Software Implementation

Firmware modules are organized into sensor acquisition, calibration/filtering, control logic, actuator control, TCP/IP communication, telemetry formatting, command handling, and fault management. The controller periodically samples the sensor, updates the state machine, publishes telemetry, and checks watchdog/fault conditions. The dashboard presents current values and allows authorized commands.

Modern Tools Used

Tool / Software / Equipment	Purpose	Stage Used
Arduino IDE / PlatformIO	Firmware development and serial debugging	Development
ESP32 Wi-Fi/TCP-IP stack	Network communication	Integration
Python or browser dashboard stack	Telemetry visualization and test client	Integration / testing

Serial monitor / logic tools	Timing and message debugging	Verification
Git (recommended)	Version control and traceability	Development
Multimeter	Voltage/current continuity checks	Hardware verification

Implementation evidence such as prototype photographs, wiring diagrams, serial logs, dashboard screenshots and final measured data should be inserted here before final institutional submission.

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
Soil moisture sensing	Compare sensor response in dry, reference-moist and wet conditions	Monotonic response and repeatable mapping; calibration documented
Sampling interval	Timestamp successive readings	Measured interval within configured tolerance
Automatic ON	Reduce soil moisture below T_ON	Pump/valve transitions ON once threshold condition is met
Automatic OFF	Increase moisture above T_OFF	Pump/valve transitions OFF
Hysteresis	Oscillate input around threshold	No rapid repeated switching
TCP/IP telemetry	Capture packets/client messages on LAN	Current moisture and state received without malformed messages
Remote command	Send authorized and unauthorized commands	Authorized command accepted; unauthorized command rejected
Network loss	Disable network during operation	Local safety/control continues; remote state marked unavailable
Sensor fault	Disconnect or force invalid sensor reading	Automatic irrigation inhibited and fault indicated
Actuator timeout	Simulate/allow maximum run time	Pump is stopped at configured safety limit

4.4 Results

No experimental dataset was supplied with the project brief. Therefore, this section distinguishes engineering acceptance targets from actual achieved measurements. The table below is a validation template with example target values; these values must be replaced with the team's measured results before claiming completed experimental performance.

Specification	Required Value	Achieved Value	Status
Sampling interval	10 s configured	To be measured	Pending
Moisture state update	≤10 s under normal operation	To be measured	Pending
Automatic threshold action	Correct ON/OFF state	To be measured	Pending
Hysteresis stability	No chatter for stable	To be measured	Pending

	input		
TCP/IP telemetry	Valid telemetry on LAN	To be measured	Pending
Remote command authorization	Unauthorized command rejected	To be measured	Pending
Network-loss behavior	Local control remains active	To be measured	Pending
Sensor-fault behavior	Automatic watering inhibited	To be measured	Pending
Maximum pump runtime	Stops at configured timeout	To be measured	Pending

Recommended measured result plots include: (1) soil-moisture percentage versus time during drying/wetting, (2) pump state versus soil moisture, (3) TCP/IP message latency or update interval, and (4) system response during network-loss and sensor-fault tests.

4.5 Data Analysis & Engineering Judgment

The principal engineering metrics are sensing repeatability, control stability, communication continuity, and safe fault response. Sensor accuracy should be reported only against a defined reference method because a raw sensor percentage is not an absolute measure of volumetric water content unless calibrated for the soil and installation. Network performance should be evaluated under the intended LAN conditions and should distinguish application update interval from raw packet latency.

A successful system should show a clear relationship between moisture state and actuation: moisture below the lower threshold should cause irrigation, while moisture above the upper threshold should stop irrigation. If repeated ON/OFF transitions occur near a threshold, the engineering response is to increase hysteresis, add filtering, or both. If network outages stop local irrigation control, the architecture should be revised so that network communication remains supervisory rather than safety-critical.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement (current report status)
Design complete architecture	Fig. 3.1 and design specification	Fully defined
Develop sensing interface	Sensor/controller interface specified	Implementation evidence required
Implement hysteresis control	Control equation and state logic defined	Implementation evidence required
TCP/IP communication	Network architecture and requirements defined	Implementation evidence required
Remote monitoring/management	Dashboard requirements defined	Implementation evidence required
Testing and validation	Test plan and acceptance criteria defined	Measured data required
Evaluate safety/security/sustainability	Tables 3.7 and risk register	Fully addressed at design level

4.7 Strengths & Limitations

Strengths:

- Closed-loop moisture feedback rather than timer-only irrigation.
- Local control continues to operate when the remote network is temporarily unavailable.
- TCP/IP connectivity supports remote monitoring and management.
- Hysteresis and runtime limits reduce actuator stress and unsafe continuous operation.
- Modular architecture supports future multi-zone expansion.

Limitations:

- Moisture readings require soil-specific calibration and can be affected by sensor placement.
- Wi-Fi/TCP-IP availability is not guaranteed in all agricultural environments.
- The prototype does not by itself detect pipe leakage or verify actual water flow.
- Weather, crop type, evapotranspiration and soil hydraulic properties are not yet incorporated into the decision model.
- Field durability and long-term sensor drift require extended outdoor testing.
- Formal certification and production-grade cybersecurity assessment are outside the prototype scope.

4.8 Project Planning, Roles & Budget

Milestone	Planned Activity
Week 1	Problem definition, requirements and literature review
Week 2	Architecture and component selection
Week 3	Sensor interface and calibration prototype
Week 4	Actuator driver and safety checks
Week 5	Local control firmware and hysteresis
Week 6	TCP/IP communication and telemetry
Week 7	Remote dashboard and command authorization
Week 8	System integration and fault testing
Week 9	Data collection, analysis and refinement
Week 10	Documentation, review and final demonstration

Student	Assigned Responsibility	Actual Contribution
Student 1	System architecture and embedded firmware	Update with actual evidence
Student 2	Sensors and irrigation hardware	Update with actual evidence
Student 3	TCP/IP and dashboard	Update with actual evidence
Student 4	Testing, documentation and coordination	Update with actual evidence

Item	Estimated Cost (INR)	Actual Cost (INR)
ESP32 development board	500	To be entered

Capacitive soil-moisture sensor	250	To be entered
Relay/MOSFET driver	150	To be entered
DC pump / solenoid valve	600	To be entered
Power supply and protection	500	To be entered
Tubing, connectors, wires and enclosure	700	To be entered
Prototype PCB/breadboard	300	To be entered
Miscellaneous	500	To be entered
Total	3500	To be entered

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

This project addresses the engineering problem of making irrigation more responsive to actual soil condition while enabling remote farm supervision. The proposed solution integrates soil-moisture sensing, embedded threshold/hysteresis control, safe pump or valve actuation, and TCP/IP communication within a single modular architecture.

The design emphasizes a key system-level principle: remote networking should enhance irrigation management without becoming a single point of failure for local control. The controller therefore performs the essential sensing and safety logic locally, while the TCP/IP layer provides telemetry and authorized remote commands. The report also defines requirements, standards/guidance, trade-offs, sustainability considerations, security measures, risks, a verification plan and project-management structure.

The architecture is technically suitable as a capstone prototype, but final claims of accuracy, latency, reliability or water savings require measured experimental data. Once those measurements are collected, the same acceptance framework can be used to quantify performance and guide refinement.

5.2 Future Work

- Deploy multiple irrigation zones with independent moisture sensors and valves.
- Add water-flow and pressure sensors to detect leaks, blocked lines and pump failure.
- Integrate weather forecasts, rainfall sensing and evapotranspiration estimates.
- Replace fixed thresholds with adaptive or predictive irrigation algorithms.
- Store long-term telemetry for trend analysis and crop-cycle comparison.
- Use secure TLS-based communication and device certificates for production deployment.
- Add local non-volatile configuration and secure firmware-update mechanisms.
- Evaluate solar power and low-power operation for remote field installations.
- Conduct multi-season field trials across different soil types and crops.
- Perform formal electrical, environmental and cybersecurity compliance assessments.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

- Embedded sensor interfacing and calibration concepts.
- TCP/IP-based IoT communication and remote command handling.
- Feedback-control concepts such as thresholds, hysteresis and state machines.
- Electrical load interfacing, isolation and protection for pump/valve control.
- Engineering verification, risk assessment and technical documentation.

Engineering & Professional Skills Developed:

- System architecture and interface definition.
- Embedded programming and debugging.
- Network troubleshooting and protocol-level reasoning.
- Test planning, acceptance criteria and engineering judgement.
- Team coordination, documentation and responsible reporting.

Student	**Individual Reflection (to be personalized before**

	submission)
Student 1	The most difficult task was integrating sensing, control and communication without allowing network problems to compromise local operation. The key decision was to keep irrigation control on the controller and use TCP/IP primarily for supervision. Independent learning focused on embedded networking and fault-handling techniques. If repeated, the design would include earlier integration testing and a more formal configuration-management process.
Student 2	The major challenge was creating a reliable interface between low-voltage control electronics and the irrigation actuator. Component ratings, isolation and safe switching were treated as primary design constraints. New learning included sensor calibration and load-protection concepts. If repeated, the prototype would include flow feedback and a more durable outdoor enclosure from the beginning.
Student 3	The most difficult engineering issue was designing a useful remote-management interface while keeping commands controlled and traceable. The selected TCP/IP approach provides a straightforward basis for telemetry and future scaling. Independent learning focused on client-server communication and IoT security practices. If repeated, secure transport and device authentication would be incorporated earlier.
Student 4	The main challenge was converting the project idea into measurable requirements and verification tests. A key decision was to separate acceptance targets from actual results so that unsupported performance claims are avoided. New learning included engineering risk registers and technical reporting. If repeated, more time would be allocated for controlled field experiments and statistical analysis.

REFERENCES

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[5] IETF, “Transmission Control Protocol (TCP),” RFC 9293, Internet Engineering Task Force, 2022.

[6] IETF, “HTTP Semantics,” RFC 9110, Internet Engineering Task Force, 2022.

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[10] OWASP Foundation, “OWASP Internet of Things Project,” security guidance for IoT devices and deployments.

[11] ISO/IEC, “ISO/IEC 27001: Information security, cybersecurity and privacy protection — Information security management systems — Requirements,” International Organization for Standardization.

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[13] FAO, “The State of the World’s Land and Water Resources for Food and Agriculture,” Food and Agriculture Organization of the United Nations.

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[15] Manufacturer datasheets for the selected soil-moisture sensor, relay/driver, pump/valve and power supply used in the final prototype.

Note: Before final submission, replace generic manufacturer entries with the exact part numbers, datasheet revisions, URLs/access dates, datasets and software versions actually used by the team. Any AI-assisted resources should also be acknowledged according to institutional policy.

APPENDICES

Appendix A – Detailed Calculations

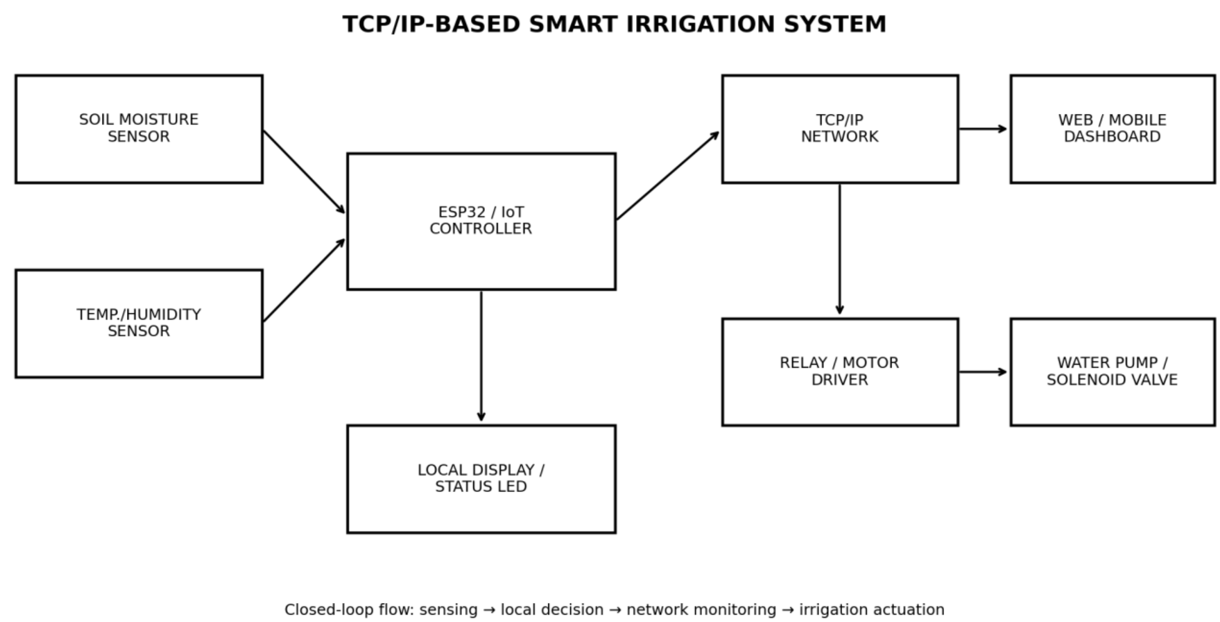
A.1 Pump power estimate: $P = V \times I$. Use the measured operating voltage and current of the selected pump to calculate input power. Include startup current where relevant for driver and supply selection.

A.2 Controller energy estimate: $E_{day} \approx P_{controller} \times 24$ h for continuous operation. For battery/solar designs, include duty cycle, conversion losses and battery reserve.

A.3 Hysteresis logic: $T_{ON} < T_{OFF}$. The difference $T_{OFF} - T_{ON}$ is the hysteresis band and should be selected using observed sensor noise and crop/soil response requirements.

Appendix B – Engineering Drawings / Schematics

Insert the final wiring schematic showing sensor connections, controller pins, power rails, actuator driver, flyback/protection components, isolation boundary and connector labels.



Appendix Figure B.1: Reference system architecture

Appendix C – Source Code / Code Repository Information

Insert the approved repository URL or attach the relevant firmware excerpts. Recommended module structure: sensor.cpp/h, control.cpp/h, actuator.cpp/h, network.cpp/h, dashboard/client code, configuration.h, and main application.

Appendix D – Datasheets

Attach datasheets for the exact ESP32 board, soil-moisture sensor, relay/MOSFET driver, pump/valve, power supply, enclosure and protection components used.

Appendix E – Experimental / Raw Data

Attach timestamped CSV/log files containing soil-moisture readings, actuator state, network events, sensor-fault events and test identifiers. Record soil type, sensor depth, sampling interval, environmental conditions and calibration procedure.

Appendix F – Ethics / Institutional Approval

Include institutional approvals only if the project involves human participants, protected data, regulated trials or other activities requiring approval. Otherwise state that no such approval was applicable.

Appendix G – Project Meeting / Progress Records

Insert dated meeting records, task assignments, decisions, design-review outcomes and supervisor feedback.

Appendix H – User / Industry Feedback

Attach feedback from farmers, supervisors, technicians or other relevant stakeholders, where applicable.

Appendix I – Publications / Patents / Competitions / Product Outcomes

List any publications, patent applications, competition participation or product outcomes only if they actually occurred.

Appendix J – Individual Contribution Evidence

Attach commit history, laboratory records, photographs, test sheets, design files or other evidence supporting each student's contribution.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

Outcome	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1 and Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3.7
Teamwork and leadership	SO5	Ch. 4.8 + Contribution Statement
Experimentation, analysis, engineering judgment	SO6	Ch. 4.3–4.6
Independent acquisition of new knowledge	SO7	Ch. 5.3
Standards	SO2	Ch. 3.2
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3.7
Risk assessment and mitigation	SO2/SO4	Ch. 3.7
Security considerations	Relevant SO	Ch. 3.7
Integrated/system-level engineering	SO1/SO2	Ch. 3.5
Project planning and management	SO5	Ch. 4.8
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4.8

Department/faculty assessor: complete the final assessment fields as required by the institution.